ENV S 193DS Final

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Github Repo

Set up

```
# reading in packages
library(here)
library(tidyverse)
library(janitor)
library(dplyr)
library(MuMIn)
library(DHARMa)
library(ggeffects)
library(ggplot2)
library(readxl)

# reading in data
nests <- read.csv(here("data/nest_data_final.csv"))
my_data <- read_xlsx(here("data/my_data.xlsx"))
```

Problem 1. Research Writing

a. Transparent statistical methods

In part 1, they used a logistic regression because they are calculating a prediction or probability of how one variable effects the other, and the response variable, American Avocet microhabitat use, is a binary variable.

In part 2, they used a Chi square test because both predictor and response variables are categorical.

b. Suggestions for rewriting

Part 1: American Avocet microhabitat use depends on distance to water edge.

Part 2: Recorded bird species (American Avocet, Forster's Tern, Black-necked Stilt), differ significantly based on habitat type(managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is which is proximity to island nesting sites (m) because islands are important waterbirds for nesting and breeding as shown in several studies (Schacter C. et al. 2023 introduction paragraph 2 page 3).

Problem 2. Data analysis

a. Clean and wrangle

```
# new object nests_clean
nests_clean <- nests |>
  # keeping only 3 variables
  # variables: tree height, nest occurrence, and ant species
  select(height_cm, case_control, ant_species) |>
  # only including ant species AB, RRB, and BBR
  filter(ant_species %in% c(
                     "AB",
                   "RRB",
                   "BBR")) |>
  # reordering ant species (AB,RRB, BBR)
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                    "AB",
                                    "RRB",
                                    "BBR")) |>
  # creating new row with full species names
  mutate(species_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  ))

# displaying random ten rows of data
slice_sample(nests_clean, n = 10)
```

```
   height_cm case_control ant_species            species_name
1      169.0            0         RRB   Crematogaster mimosae
```

```
2       144.0             0          RRB   Crematogaster mimosae
3       200.5             0          RRB   Crematogaster mimosae
4       165.5             1          BBR Crematogaster nigriceps
5       248.5             1          RRB   Crematogaster mimosae
6       390.0             1          BBR Crematogaster nigriceps
7       199.0             0          RRB   Crematogaster mimosae
8       234.5             0          RRB   Crematogaster mimosae
9       114.5             0          RRB   Crematogaster mimosae
10      154.0             0          RRB   Crematogaster mimosae
```

b. Response variable

For the response variable, case_control, the 1's signify there is a nest, of any bird species, present, while a 0 signifies there is no nest present.

c. Purpose of study

Tree height: Tree height is a continuous variable. Here, as tree height increases, the probability of a nest occurrence could increase because of increased distance between nests and browsing mammals (introduction section paragraph 3).

Acacia ant species: Acacia ant species is a categorical variable. Here, C. mimosoe and C. nigriceps could have a higher probability nest occurrence because they aggressively defend trees against mammalian and insect herbivores (introduction section paragraph 2).

d. Table of models

Model number	Tree Height	Ant Species	Interaction
1	X	X	X
2	X	X	

e. Run the models

```
# model 1: Model with interaction term

# creating new object model1 - interaction term
# response ~ predictor * predictor
model1 <- glm(case_control ~ height_cm * ant_species,
```

```r
              data = nests_clean,
              # binomial/logistic regression
              family = "binomial"
              )

# model 2: Model without interaction term

# creating new object model2 - no interaction term
# response ~ predictor + predictor
model2 <- glm(case_control ~ height_cm + ant_species,
              data = nests_clean,
              # binomial/logistic regression
              family = "binomial"
              )
```

f. Select the best model

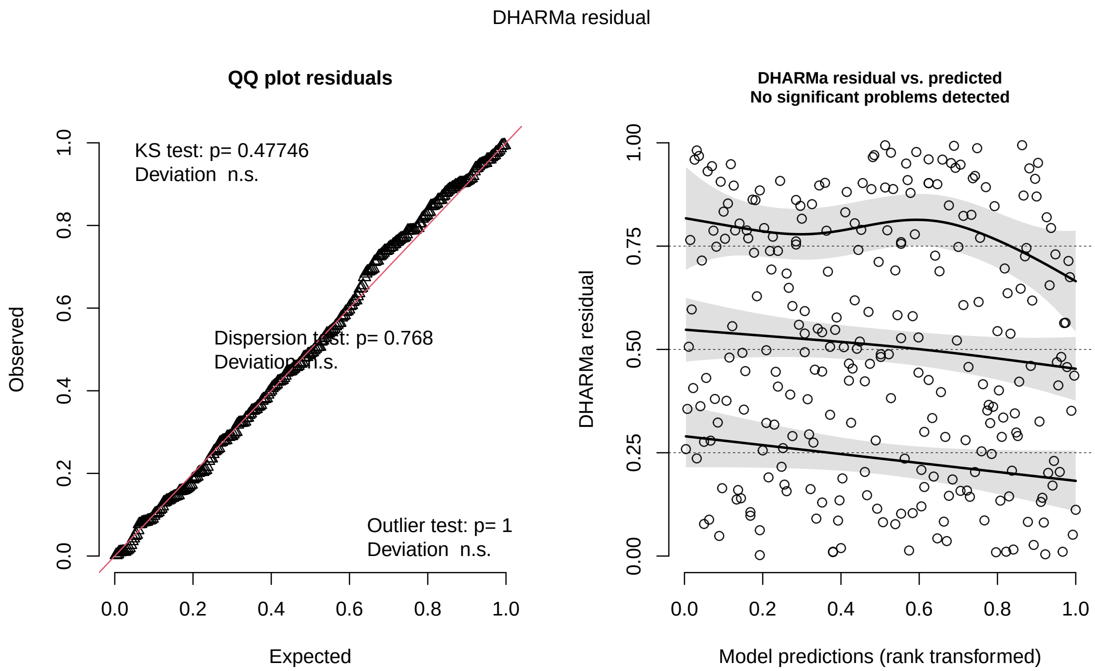
```r
# AKaike's Information Criterion
# listing models
AICc(
model1,
model2
) |>
# asking to order/display rank of models
  arrange(AICc)
```

```
       df     AICc
model1  6 238.1236
model2  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) was model 1, with interaction.

g. Check the diagnostics

```r
# Displaying diagnostics for model of best fit (model1)
plot(
  simulateResiduals(model1)
)
```

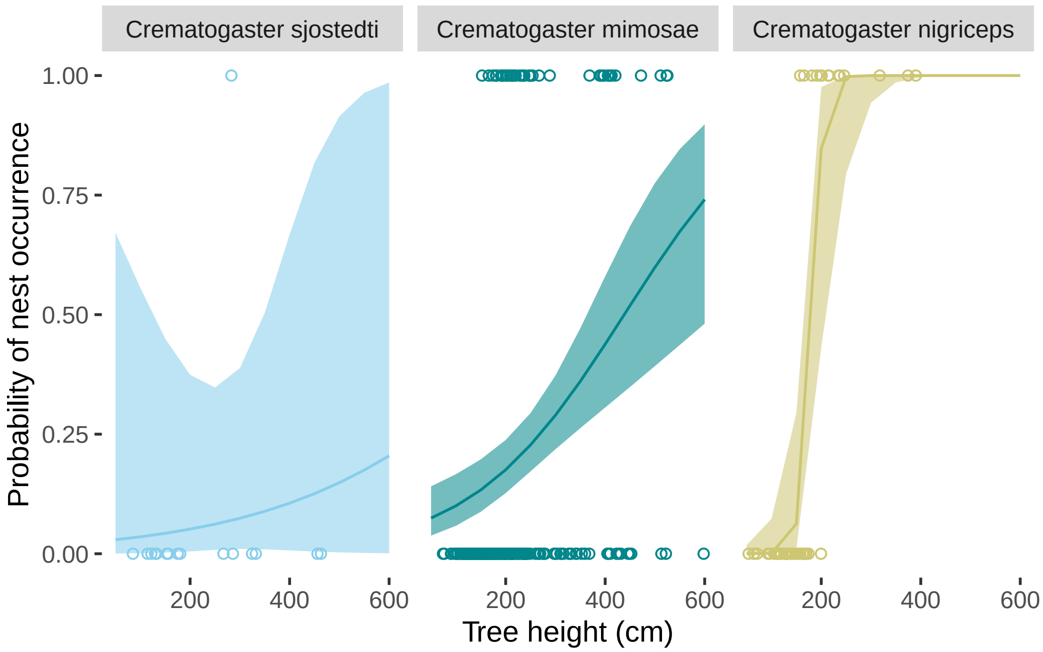


h. Visualize the model predictions

```
# creating new object mod_preds
mod_preds <- ggpredict(model1,
                       #adding predictor variables
                       terms = c("height_cm",
                                 "ant_species")) |>
  #renaming group column ant_species
  dplyr::rename(ant_species = group)


#ggplot base
# x = height, y = nest occurrence, categorized by ant species
ggplot(nests_clean,
       aes( x = height_cm,
            y = case_control)) +
  # adding points representing observations
  geom_point(aes(color = ant_species),
             shape = 21) +
  # adding confidence level ribbon based on mod_preds
```

```
  geom_ribbon(data = mod_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = ant_species,
              alpha = 0.1)) +
  # adding prediction line
  geom_line(data = mod_preds,
            aes( x = x,
                 y = predicted,
                 color = ant_species)) +
  # creating different panels
  # labelling by full name instead of abbreviations
  facet_wrap(~ant_species,
             labeller = labeller(
               ant_species = c(AB = "Crematogaster sjostedti",
                               RRB = "Crematogaster mimosae",
                               BBR = "Crematogaster nigriceps"))) +
  #custom colors
  scale_color_manual(values = c(
    "AB" = "skyblue",
    "RRB" = "turquoise4",
    "BBR" = "khaki3"
  )) +
  # matching ribbon colors
  scale_fill_manual(values = c(
  "AB" = "skyblue",
  "RRB" = "turquoise4",
  "BBR" = "khaki3"
)) +
  # adding x range
  # adding line breaks
  scale_x_continuous(limits = c(50, 600),
                     breaks = c(200, 400, 600)) +
  # getting rid of legend
  theme(legend.position = "none",
        panel.background = element_blank(), # removing grey background
        panel.grid.major = element_blank() # removing grid lines
        ) +
 # labelling axes
   labs(x = "Tree height (cm)",
```

```
     y = "Probability of nest occurrence")
```



i. Write a caption for your figure

Figure 1: As height increases, the probability of bird nest occurrence increases with each acacia ant species. Data from nest_data_final.csv (Lujan E.et al, Zenodo, 2023). Lines represent predictions of probability of nest occurrence (lbs) given acacia species; Crematogaster sjostedti (AB), Crematogaster mimosae (RRB), Crematogaster nigriceps (BBR), (total n = 270) and tree height in cm. Meanwhile, points represent actual observations of if a nest occurred or not(0 = nest not present, 1 = nest present). The probability of nest occurrence increases as tree height increases. Nevertheless, the probability increases at different rates with each ant species, where Crematogaster sjostedti has the most gradual increase in probability followed by Crematogaster mimosae and Crematogaster nigriceps. Ribbons represent a 95% confidence interval. Colors represent acacia ant species (skyblue: Crematogaster sjostedti, teal: Crematogaster mimosae, khaki: Crematogaster nigriceps).

j. Calculate model predictions

```r
# predicting nest occurence probability at 600 cm
ggpredict(model1,
          terms = c("height_cm [600]",
                    "ant_species"))
```

```
# Predicted probabilities of case_control

ant_species: AB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

Acacia ant species and tree height (cm) interact to predict the probability of nest occurrence. As tree height increases, probability of nest occurrence increases.

Furthermore, at a tree height of 600 cm, the probability of nest occurrence is 0.20 for Crematogaster sjostedti (95% CI: [0.00, 0.99]), 0.74 for Crematogaster mimosae (95% CI: [0.48, 0.90]), and 1.00 for Crematogaster nigriceps (95% CI: [1.00, 1.00]). Indicating higher probability of nest occurrence for Crematogaster nigriceps at the highest tree height measured, 600 cm.

Here, the higher probability of nest occurrence for acacia ant species C. mimosae and C. nigriceps could be explained because they are more aggressive defenders agains mammalian and insect herbivores, thereby defending nests against these predators and increasing probability of occurrence(Lujan E. et al. 2023, introduction section paragraph 2).

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

There are many differences but a large difference is the amount of observations and data points, also of course they are displayed very differently. The affective visualization is a lot more artistic and its hard to tell it is data at first. Both graphs converge however in showing the low amount of caffeine consumed the more sleep in hours was had. Nevertheless, the patterns differ somewhat between the two. The graph from homework 02 did not have any distinct patterns and looked relatively random although there were a couple points that showed a pattern of negative correlation starting to emerge. The affective visualization had a much more prominent decrease and much more data to go off of. From week 9, I received mostly suggestions about coloring and incorporated these through watercolor for my final product.

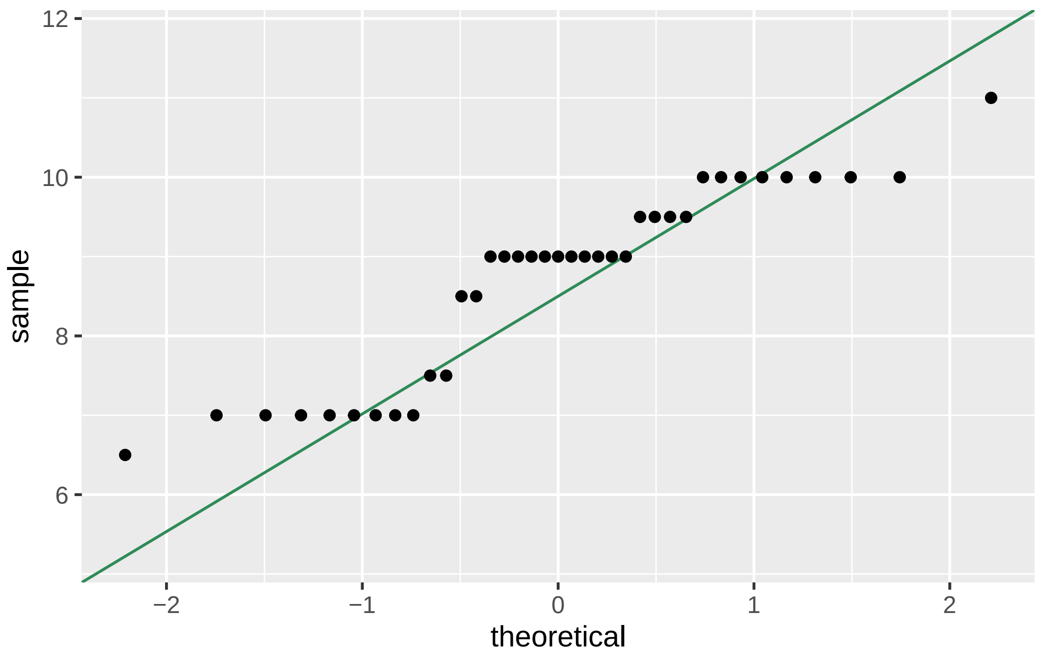
b. Designing an analysis

A Pearson's correlation could be applied to answer the research question: How does the amount of sleep in hours influence the amount of coffee consumed in ounces per day? Here, a Pearson's correlation is appropriate because it uses parametric tests measuring the mean, there are no significant outliers in the data, and the research question asks for the strength of correlation or how strongly the predictor influences the response variable.

c. Check any assumptions and run your analysis

Checking assumptions

```r
# new object, my_data clean
# cleaning my_data
my_data_clean <- clean_names(my_data)

# QQ plot to check assumptions
#base layer: my_data_clean
ggplot(data = my_data_clean,
       #y axis: amount of sleep
       mapping = aes(sample = amount_of_sleep)) +
  #first layer: reference for qq plot
  geom_qq_line(color = "seagreen") +
  #second layer: points representing sample quantiles
  geom_qq()
```



Running Pearson's Correlation

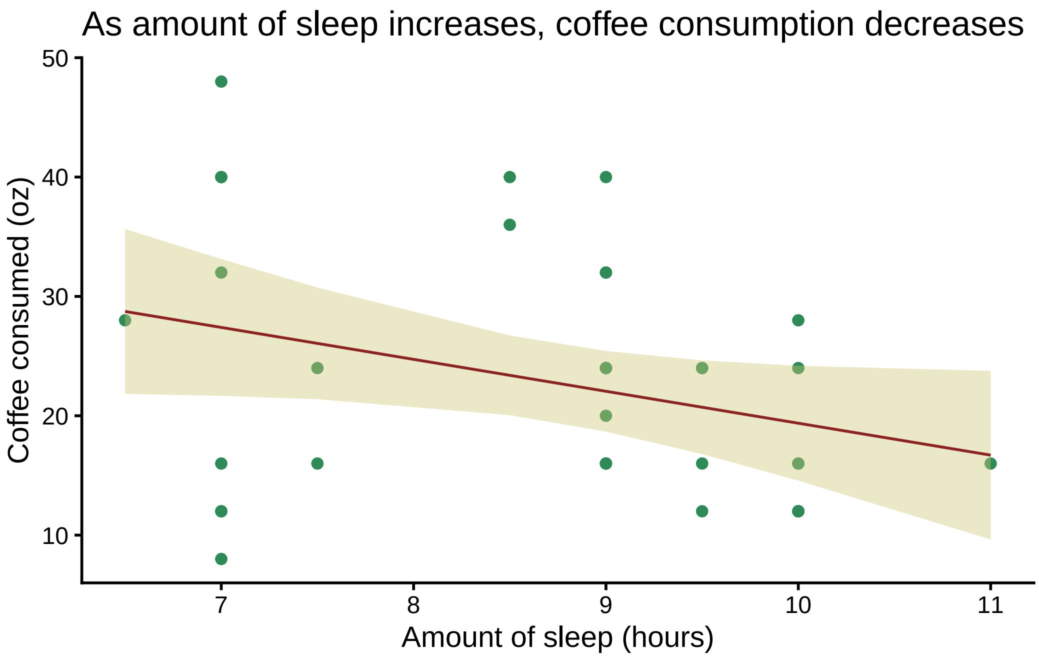
```
# running pearsons correlation
# data$predictor, data$response
cor.test(my_data_clean$amount_of_sleep, my_data_clean$coffee_oz,
         method = "pearson")
```

```

    Pearson's product-moment correlation

data:  my_data_clean$amount_of_sleep and my_data_clean$coffee_oz
t = -1.9835, df = 35, p-value = 0.0552
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.581958286  0.006843215
sample estimates:
       cor
-0.3178806
```

d. Create a visualization

```
# creating new object
# linear regression model of my data
# (response ~ predictor)
my_data_lm <- lm( coffee_oz ~ amount_of_sleep,
    data = my_data_clean
    )

# creating new object, my_data_preds
my_data_preds<- ggpredict(my_data_lm,
                         terms = "amount_of_sleep")
# base layer ggplot
# x = amount of sleep
# y = coffee consumed
ggplot(my_data_clean,
       aes(x = amount_of_sleep,
           y = coffee_oz)) +
  # adding my data in points
  geom_point(color = "seagreen") +
  # adding confidence interval ribbon
  geom_ribbon(data = my_data_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              # ribbon color
              fill = "khaki3",
              alpha = 0.4) +
  # adding fitted linear model
  geom_line(data = my_data_preds,
            aes( x = x,
                 y = predicted),
            # line color
            color = "brown4") +
  # labelling axes and title
  labs(
    x = "Amount of sleep (hours)",
    y = "Coffee consumed (oz)",
    title = "As amount of sleep increases, coffee consumption decreases"
  ) +
  # changing theme from default
```

```
  theme_classic() +
  # getting rid of legend
  theme(legend.position = "none")
```



e. Write a caption

Figure 2: As amount of sleep increases, coffee consumption decreases. Data from my_data.xlsx (van der Poel, 2026). Line represents predicted values following a linear model, demonstrating relationship between amount of sleep (hours) and coffee consumed (oz) (total n = 37). Meanwhile, points represent actual observations of coffee consumed on a given day given a certain amount of sleep. The amount of coffee consumed decreases as amount of sleep increases. Ribbon represents a 95% confidence interval. Colors represent different graph elements (brown: fitted line, seagreen: observations, khaki: confidence interval ribbon).

f. Write about your results

As amount of sleep increases (hours), amount of coffee consumed decreases(oz) (total n = 37). After running a Pearsons correlation test, I found a weak negative relationship between amount of sleep and coffee consumed (Pearson's r = 0.32, t(35) = -2.0, p = 0.055, α = 0.05).

g. Sharing your affective visualization